

ABSTRACT

This project focuses on the design and development of an intelligent voice-controlled cybernetic surveillance rover aimed at enhancing security, automation, and human-machine interaction in smart environments. Traditional surveillance systems are mostly static and require continuous human monitoring, limiting their efficiency and responsiveness. To address these challenges, the proposed system introduces a mobile robotic platform integrated with artificial intelligence, enabling real-time surveillance, autonomous navigation, and interactive control through natural language commands.

A key feature of the system is its voice-first control mechanism, allowing users to operate the rover using simple commands such as movement, scanning, and status checks. The system incorporates advanced AI models for natural language understanding and decision-making, enabling it to interpret user intentions and respond intelligently. Additionally, computer vision techniques are used to process live video streams from an onboard camera, providing real-time monitoring and visual feedback through an integrated display interface.

The rover is equipped with a multi-sensor obstacle detection system using infrared sensors, ensuring safe and collision-free navigation. A differential drive mechanism controlled through PWM signals enables smooth and precise movement. Furthermore, the system includes an IoT-based home automation module, allowing the rover to control electrical appliances via a relay interface.

Designed for applications in smart homes, offices, and industrial environments, this system offers a scalable and efficient solution for modern surveillance needs. By integrating AI, real-time sensing, and embedded control, the project delivers a user-friendly and autonomous robotic platform that enhances security, convenience, and operational efficiency.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
A	Ampere
AC	Alternating Current
ADC	Analog to Digital Converter
AI	Artificial Intelligence
API	Application Programming Interface
CPU	Central Processing Unit
DC	Direct Current
ESP32	Espressif 32-bit Microcontroller
GND	Ground
GPIO	General Purpose Input/Output
GOKU	Groq-integrated Operational Kinetic Unit
GPU	Graphics Processing Unit
HMI	Human-Machine Interface
Hz	Hertz
IR	Infrared
IoT	Internet of Things
LCD	Liquid Crystal Display
LLM	Large Language Model
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical Systems
NLP	Natural Language Processing
PCB	Printed Circuit Board

Abbreviation	Full Form
PWM	Pulse Width Modulation
RL	Reinforcement Learning
SLAM	Simultaneous Localization and Mapping
SMS	Short Message Service
UI	User Interface
V	Voltage
VCC	Voltage Common Collector
WiFi	Wireless Fidelity